

EXERCISES ON ADJUNCTIONS

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Notation. We denote an adjunction by an opposing pair of arrows: $F: C \rightleftarrows D : G$. In this notation, F will always be the left adjoint and G the right adjoint.

Exercise 1. Let $F: C \rightleftarrows D : G$ be an adjunction. Prove that the right adjoint G is fully faithful if and only if the counit $\varepsilon: FG \Rightarrow \text{id}_D$ is a natural isomorphism. Notice that in this case the left adjoint F is essentially surjective on objects. Dually, F is fully faithful if and only if the unit $\eta: \text{id}_C \Rightarrow GF$ is a natural isomorphism.